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Patent Public Search | Text View

United States Patent Application Publication

20250282279

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

OLSON; Justin et al.

QUICK HOLD AND RELEASE CLAMP AND METHOD OF RELEASABLY JOINING A FIRST AND SECOND BODY USING SAME

Abstract

A releasable clamp, a method for releasably joining a first body to a second body and an actuator are provided. A clamp may be insertable through aligned openings in joinable bodies. The clamp may comprise a base comprising a compression surface facing the first body and being compressible around a periphery of the first opening, and a protrusion extending in a longitudinal direction from the base and insertable through the aligned openings. A force may be applied to an actuator operatively connected to a retractable pawl or pawls housed in the base and extendable in a lateral direction perpendicular to the longitudinal direction. When the pawl or pawls are extended to an engaged position, the second body, the first body and the base are held into compressed relation against the compression surface. A sloped surface on the pawl or pawls allows the clamp to secure joinable bodies of various thicknesses.

Inventors: OLSON; Justin (Lincoln, NE), BUTLER; Brian (Lincoln, NE), TAKAHASHI; Keiji (Lincoln, NE), BOUDREAU; Dominique (Drummondville, CA), BESSETTE; Frédéric (Drummondville, CA), LECLERC; Jean-Michel (Drummondville, CA), POUDRIER; Alexandre (Drummondville, CA)

Applicant: KAWASAKI MOTORS, LTD. (Akashi-shi, JP); SOUCY INTERNATIONAL Inc. (Drummondville, CA)

Family ID: 1000008105322

Assignee: KAWASAKI MOTORS, LTD. (Akashi-shi, JP); SOUCY INTERNATIONAL Inc. (Drummondville, CA)

Appl. No.: 18/798859

Filed: August 09, 2024

Related U.S. Application Data

Publication Classification

Int. Cl.: **B60P7/08** (20060101); **F16B2/18** (20060101)

U.S. Cl.:

CPC **B60P7/0807** (20130101); **F16B2/18** (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a releasable clamp by which a first body such as an accessory body can be quickly attached to and released from a second body, such as a vehicle body.

BACKGROUND ART

[0002] Recreational and work vehicles continue to grow in popularity for transporting people to sometimes remote destinations over rugged terrain for the purpose of engaging in outdoor leisure or work activities. Quite often, these activities involve an extensive range of equipment, such as bicycling equipment, camping equipment, hunting equipment, and/or work equipment. Such equipment may be carried in containers or racks, or other structures referred to herein generally as “accessories,” in the vehicle.

[0003] In order to ensure that these accessories remain attached to the vehicle, a means is required to secure these accessories to the vehicle. However, difficulties may arise in that depending on the manufacturer of the accessories and/or of the vehicle, the bodies of the accessories and the vehicle may have widely varying thicknesses, and no clear position in which to place the mounting mechanisms, making it difficult to create a mechanism that is widely adaptable across a wide range of accessory and vehicle bodies thicknesses.

[0004] Various mechanism designs have been attempted such as screw-type fasteners, (see Patent Document 1: U.S. Patent Application Publication No. 2011/0020092); and snap type fasteners (see Patent Document 2: U.S. Patent Application Publication No. 2013/0287517 and Patent Document 3: U.S. Patent Application Publication No. 2017/0335877) as examples. However, conventional approaches have suffered drawbacks in that they are difficult to deploy, lack the strength to accommodate heavier accessories, are prone to breakage, are subject to slippage or engagement failure, or fail to provide a secure engagement between the body of the vehicle and the body of the accessory (e.g. to prevent rattling or vibration), and other drawbacks attributable, at least in part, to the inability to meaningfully accommodate different body thicknesses of the accessory and the vehicle bodies.

[0005] For example, in Patent Document 1, a panel fastener is provided with an actuator pin and a retainer mechanism. The fastener arrangement in Patent Document 1 requires a tool to turn knob **10**, or knob **10** may be rotated by hand in order to rotate a pin **11** to engage flange **16** at a fixed depth. However, according to the arrangement in Patent Document 1, it may be difficult to accommodate panels having varying thicknesses.

[0006] In the two-piece clip described in Patent Document 2, an engagement convex portion may include a pin that is pushed into a grommet of a fixed length body portion to a maximum push-in position that enables the sides of the body portion to engage the walls of an insertion hole. However, according to the arrangement in Patent Document 2, it may be difficult to accommodate different depths in the bodies in which a hole is formed. Further, since the body is made of resin, it

is likely to be easily breakable when exposed to shear forces that may be experienced in a vehicular application. The fastening clip assembly described in Patent Document 3 may suffer similar drawbacks.

SUMMARY

[0007] In view of the above and other drawbacks of conventional arrangements, one or more embodiments are provided for a quick hold and release clamp and a method of releasably joining a first body to a second body using a clamp.

[0008] In an embodiment or embodiments, a clamp may be provided for releasably joining a first body to a second body. The clamp may be insertable through a first opening in the first body and a second opening in the second body aligned with the first opening. The clamp may comprise: a base comprising a compression surface that faces the first body and is compressible around a periphery of the first opening, and a protrusion extending in a longitudinal direction from the base, the protrusion insertable through the first and second openings and connectable to the second body; an actuator connected to the base; and a retractable pawl assembly housed in the base, the pawl assembly having a pawl or pawls extendable from the protrusion of the base in a lateral direction perpendicular to the longitudinal direction, the pawl assembly operatively coupled to the actuator. In an embodiment or embodiments, the pawl or pawls may be retracted into the protrusion in a disengaged position by operation of the actuator through a force applied to the actuator. The pawl or pawls may be extended from the protrusion of the base to an engaged position to engage the second body and to hold the second body, the first body and the base into compressed relation against the compression surface by force applied to the actuator. It will be appreciated that the pawl or pawls may extend from and retract into the protrusion in a direction orthogonal to the rotation axis, e.g. may be retracted to release the engagement. Each of the pawl or pawls may comprise a cam surface that slopes to widen in a direction away from the base to variably engage with the second body. The cam surface may extend to variably engage the second body to hold the second body, the first body and the base into compressed relation against the compression surface, e.g. despite variations in thicknesses between bodies being joined.

[0009] The base may further comprise resilient members coupled to the actuator at one end and extending laterally from a central axis of the base in the longitudinal direction, the resilient members pressing against the pawl or pawls at an other end to urge the pawl or pawls to extend laterally outwardly by operation of the actuator.

[0010] In an embodiment or embodiments, a clamp for releasably joining a first body to a second body may be provided. The clamp may be insertable through a first opening in the first body and a second opening in the second body aligned with the first opening. The clamp may comprise a base having a compression surface; a protrusion projecting from the base in a longitudinal direction perpendicular to the compression surface relative to the base; a retractable pawl that can be switched between a first state of being housed in the protrusion and a second state of extending from the protrusion in a lateral direction perpendicular to the longitudinal direction; and an actuator that transmits power to the retractable pawl for switching the state of the retractable pawl from one of the first and second states to the other one of the first and second states in response to an operation force. The retractable pawl may comprises a surface facing the base and that slopes away from the compression surface in the longitudinal direction, and the surface of the pawl widens in a lateral direction away from the base to engage with the opening of the second body to hold the second body against the first body, the first body and the base into compressed relation against the compression surface.

[0011] The base may further comprise a resilient member operatively coupled to the pawl at one end and extending laterally from the protrusion to press against the pawl at an other end to urge the pawl outwardly by the operational force. The laterally retractable pawl extends laterally from and retracts laterally into the protrusion in a direction orthogonal to the longitudinal direction.

[0012] One or more embodiments may further include a method of releasably joining a first body

to a second body using a clamp comprising: a base having a compression surface; a protrusion projecting from the base in a longitudinal direction perpendicular to the compression surface relative to the base; a retractable pawl retractable into the base and comprising a surface; and an actuator that transmits power to the retractable pawl. The method may comprise inserting, with the retractable pawl in a first state of being retracted, the protrusion through a first opening in the first body and a second opening in the second body aligned with the first opening; and switching the state of the retractable pawl to a second state of being extended from the base in a lateral direction perpendicular to the longitudinal direction in response to the power transmitted by applying an operational force applied to the actuator. In the second state, the retractable pawl extends from the base to an engaged position to engage the second body and to hold the second body, the first body and the base into compressed relation against the compression surface. The method may further comprise switching the state of the retractable pawl from the second state in which the second body, the first body and the base are held in compressed relation against the compression surface to the first state in which the second body, the first body and the base are released from compressed relation such that the clamp is removed.

[0013] One or more embodiments may further include an actuator for engaging a releasable clamp for releasably joining a first body to a second body, the releasable clamp being insertable through a first opening in the first body and a second opening in the second body aligned with the first opening, the releasable clamp comprising: a base; a protrusion; a retractable pawl extendable and retractable along an engagement axis, the retractable pawl capable of being switched between a first state of being housed in the protrusion and a second state of extending from the protrusion. The actuator may comprise: an actuator body configured to be selectively rotatable about a first axis extending longitudinally, the actuator body extending along a second axis perpendicular to the first axis, the actuator body configured to activate operation of the releasable clamp under an activation force; a mounting portion operatively coupled to the base; a coupler operatively coupled to the actuator and extending radially from the first axis; and an internal mechanism operatively coupled to the actuator and the retractable pawl, the internal mechanism configured to convert the rotational movement of the actuator under the activation force about the first axis into a translational movement of the retractable pawl along the engagement axis. The internal mechanism may comprise: a connecting member pivotally connected to the coupler about a connecting pivot defining a connecting pivot axis extending parallel to the first axis, a resilient member provided between the connecting member and the pawl to bias the pawl away from the first axis, and a guiding member slidably connected to and guiding the connecting member along the engagement axis. The actuator may be configured to transmit the activation force to the retractable pawl through the internal mechanism, the actuator may switch the state of the retractable pawl from one of a first state in which the pawl is extended and a second state in which the pawl is retracted to the other one of the first state and the second state in response to the activation force; and the retractable pawl comprises an engagement surface facing the base and that slopes away in the longitudinal direction from a compression surface of the base. The engagement surface of the retractable pawl widens in a lateral direction away from the base. When the engagement surface comes into contact with a peripheral portion of the first and second openings of the first and second bodies, respectively, the releasable clamp provides a clamping force adaptative to various thicknesses among the first and second bodies to be joined. When the activation force is applied to the actuator such that the retractable pawl is in the second state, the resilient member applies the biasing force so as to maintain an over-center effect between the coupler and the connecting member that prevents unwanted release of the releasable clamp. The actuator may be configured so as to be operable without use of a tool, e.g. the forces applied by actuator, the arrangement of the connecting member, the pawls, and the configuration of the resilient members, and so on, may be such that secure grip may be provided by application of a hand-force to the actuator.

[0014] One or more embodiments may still further provide a releasable clamp for releasably

joining a first body to a second body, the releasable clamp being insertable through a first opening in the first body and a second opening in the second body aligned with the first opening. The releasable clamp may comprise: a base; a protrusion; a retractable pawl extendable and retractable along an engagement axis, the retractable pawl capable of being switched between a first state of being housed in the protrusion and a second state of extending from the protrusion; and an actuator in accordance with various embodiments discussed above. The actuator may be configured to transmit the activation force to the retractable pawl through the internal mechanism so as to switch the state of the retractable pawl from one of the first state and the second state to the other one of the first state and the second state in response to the activation force.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0015] FIG. 1 is a perspective diagram illustrating a side view of a releasable clamp in accordance with one or more embodiments in which a pawl or pawls are in an extended state;

[0016] FIG. 2 is a perspective diagram illustrating a side view of a releasable clamp in accordance with one or more embodiments in which a pawl or pawls are being extended or retracted;

[0017] FIG. 3 is a perspective diagram illustrating a side view of a releasable clamp in accordance with one or more embodiments in which a pawl or pawls are in a retracted state;

[0018] FIG. 4 is a perspective diagram illustrating a top view of components of a base of a clamp in accordance with one or more embodiments including resilient members;

[0019] FIG. 5 is a perspective diagram illustrating an exploded view including components of a releasable clamp in accordance with one or more embodiments including a base, an actuator, pawl or pawls, resilient members and a protruding portion;

[0020] FIG. 6 is a partial perspective diagram illustrating a view of an assembly including a first body, such as an accessory, which may be mounted to a second body, such as a vehicle body, using a releasable clamp in accordance with one or more embodiments;

[0021] FIG. 7 is a partial perspective diagram illustrating view of a first opening in a first body aligned with a second opening in a second body through which a releasable clamp may be insertable in accordance with one or more embodiments;

[0022] FIG. 8 is a partial perspective diagram illustrating view of the first opening in the first body aligned with the second opening in the second body of the FIG. 7 through which a releasable clamp is inserted in accordance with one or more embodiments;

[0023] FIG. 9A is a perspective diagram illustrating a side view of a releasable clamp in a second state in which a pawl or pawls are extended in accordance with one or more embodiments;

[0024] FIG. 9B is a diagram illustrating a top view of the releasable clamp of FIG. 9A;

[0025] FIG. 9C is a diagram illustrating a side view the releasable clamp of FIG. 9A;

[0026] FIG. 9D is a diagram illustrating an end front view of the releasable clamp of FIG. 9A;

[0027] FIG. 9E is a diagram illustrating a bottom view of the releasable clamp of FIG. 9A;

[0028] FIG. 10A is a perspective diagram illustrating a cross-sectional perspective view of the releasable clamp of FIG. 9A taken along section line AA-AA shown in FIG. 9A;

[0029] FIG. 10B is a diagram illustrating a cross-sectional view of the releasable clamp of FIG. 9A taken along section line E-E shown in FIG. 9B;

[0030] FIG. 10C is a diagram illustrating a cross-sectional view of the releasable clamp of FIG. 9A taken along section line E-E shown in FIG. 9B;

[0031] FIG. 10D is a diagram illustrating a cross-sectional view of the releasable clamp of FIG. 9A taken along section line F-F shown in FIG. 9B;

[0032] FIG. 10E is a diagram illustrating a cross-sectional view of the releasable clamp of FIG. 9A taken along section line G-G shown in FIG. 9B;

[0033] FIG. 10F is a diagram illustrating a cross-sectional view of the releasable clamp of FIG. 9A taken along section line H-H shown in FIG. 9B;

[0034] FIG. 10G is a diagram illustrating a cross-sectional view of the releasable clamp of FIG. 9A taken along section line J-J shown in FIG. 9B;

[0035] FIG. 11A is a perspective diagram illustrating a side view of a releasable clamp in an intermediate state in which a pawl or pawls are between being retracted and extended by operation of an actuator in accordance with one or more embodiments;

[0036] FIG. 11B is a diagram illustrating a top view of the releasable clamp of FIG. 11A;

[0037] FIG. 11C is a diagram illustrating a side view of the releasable clamp of FIG. 11A;

[0038] FIG. 12A is a perspective diagram illustrating a cross-sectional perspective view of the releasable clamp of FIG. 11A taken along section C-C shown in FIG. 11B;

[0039] FIG. 12B is a diagram illustrating a cross-sectional side view of the releasable clamp of FIG. 11A taken along section C-C shown in FIG. 11B;

[0040] FIG. 13A is a perspective diagram illustrating a side view of a releasable clamp in a first state in which a pawl or pawls are retracted into a protrusion by operation of an actuator in accordance with one or more embodiments;

[0041] FIG. 13B is a diagram illustrating a side top view of the releasable clamp of FIG. 13A;

[0042] FIG. 13C is a diagram illustrating a side view of the releasable clamp of FIG. 13A;

[0043] FIG. 13D is a diagram illustrating a front view of the releasable clamp of FIG. 13A;

[0044] FIG. 14A is a perspective diagram illustrating a cross-sectional perspective view of the releasable clamp of FIG. 13A taken along section line D-D shown in FIG. 13B;

[0045] FIG. 14B is a diagram illustrating a cross-sectional side view of the releasable clamp of FIG. 13A taken along section line D-D shown in FIG. 13B; and

[0046] FIG. 15 is a flow diagram illustrating a method of releasably attaching a first body to a second body with a releasable clamp in accordance with one or more embodiments.

DETAILED DESCRIPTION

[0047] Descriptions are provided herein below for one or more embodiments based on the drawings. In the respective drawings referenced herein, the same constituents are designated by the same reference numerals and a duplicate explanation concerning the same constituents is omitted for brevity. All of the drawings are provided to illustrate the respective examples only.

[0048] A perspective view of a releasable clamp **10** in accordance with one or more embodiments is shown in FIG. 1. The releasable clamp **10** may include an actuator **1**, a base **2**, a protrusion **3**, a compression surface **4** and a pawl or pawls **5**.

[0049] The various components of the releasable clamp **10** will be described in accordance with an exemplary embodiment of the releasable clamp **10** having a single instance of each of the various components (e.g. pawl **5**) for sake of brevity. It is understood that the described embodiment is illustrative and exemplary only and is not to be considered as limiting the scope of the invention. For example, more than one pawl **5** may be present and operatively connected to other components of the releasable clamp **10**, as shown in the drawings. Indeed, more or less instances of some components or features of the releasable clamp **10** may be necessary to adapt the releasable clamp **10** accordingly, as shown in the drawings.

[0050] As can be seen in FIG. 1, the protrusion **3** may protrude from the base **2** in a longitudinal direction, which may be referred to as a Z direction, assuming that the compression surface **4** is generally aligned in an X-Y plane. The longitudinal direction may correspond generally to an insertion direction of the releasable clamp into an aligned pair of openings (i.e. a vehicle opening and an accessory opening), referred to as an operating position, as will be described in greater detail hereinafter. It will be understood that practically speaking, the protrusion **3** may be aligned at various angles during insertion provided the general alignment enables insertion of the protrusion **3** into the pair of openings.

[0051] The arrangement of the releasable clamp **10** may be further understood with reference to

FIG. 2, which illustrates the actuator **1**, the base **2**, the protrusion **3**, the compression surface **4** and the pawl **5** relative to axes **A1**, **A2** and **A3**.

Actuator

[0052] The actuator **1** has an actuator body **1b** configured to be selectively rotatable about a first axis **A1**, which may be a central axis that corresponds to a longitudinal axis that extends through the center of the clamp **10**, and that extends along a second axis **A2**, which may be a lengthwise axis generally perpendicular to the first axis **A1** and generally aligned in the X-Y plane. By providing the actuator **1** that extends outwardly from the first axis **A1** along the second axis **A2**, a force may be selectively applied to the actuator **1** in order to activate operation of the clamp **10**, as will be described in greater detail below. In some embodiments, the releasable clamp **10** is configured such that the actuator **1** can be conveniently operated by an average human applying a reasonable amount of force. Stated differently, the releasable clamp **10** is intended to be a quick-release and hold clamp that does not require use of a tool to operate, but rather only application of a force to the actuator **1**.

[0053] As can be seen in the exploded view provided in FIG. 5, the actuator **1** may include a cover or top surface **1a** which may be primarily designed for ergonomical purposes and may also contain promotional or instructional information integrated thereon. The top surface **1a** may also be provided as a strengthening or structural component, such as by being made of a highly durable material or a metallic material to reduce wear thereof and consequences of contact/impact with equipment, and withstand shock from dropping and other potential use-related damage.

[0054] The top surface **1a** may also serve to prevent the unwanted disassembly of the clamp **10**. In some cases the top surface **1a** may accommodate a key-receiving portion of a locking mechanism (not shown) preventing unwanted actuation of the clamp **10**.

[0055] The actuator **1** may further include a mounting portion **1f** to which the top surface **1a** may be mounted and that may enable the actuator **1** to be coupled to the base **2**.

[0056] In addition, the actuator **1** may further include a crank portion or coupler **1c** extending radially from the first axis **A1**. As best seen in FIG. 10C, the coupler **1c** has a connecting pivot **1d** defining a connecting pivot axis **1e** extending parallel to the first axis **A1**. In some embodiments, the coupler **1c** may be a component separate from the actuator **1** and operatively connected to the actuator **1**.

Base

[0057] The base **2** has a base body **2a** defining a top surface **2b** connectable to the actuator **1** and including an opening to receive a portion of the actuator **1**, a bottom surface **2c** opposite the top surface **2b** and connectable to the protrusion **3**, and a base housing portion **2d** configured to receive at least partially some of the components of the releasable clamp **10**, as will be described in greater detail below.

Protrusion

[0058] The protrusion **3** has a protrusion body **3a** defining a top surface **3b** connectable with the bottom surface **2c** of the base **2**, a bottom surface **3c** including the compression surface **4** and a protruding portion **3d** extending outwardly along the first axis **A1** and having an opening **3e** facing the third axis **A3**, and a protrusion housing portion **3f** configured to receive at least partially some of the components of the releasable clamp **10**, as will be described in greater detail below.

Internal Mechanism

[0059] An exposed view of the protrusion **3** is provided in FIG. 4, with the base **2** and the actuator **1** being hidden for sake of clarity. Referring also to FIG. 5, broadly speaking, the actuator **1** is operatively connected to the pawl **5** via an internal mechanism **30** configured to convert the rotational movement of the actuator **1** about the first axis **A1** into a translational movement of the pawl **5** along the third axis **A3**.

[0060] The internal mechanism **30** will now be described. The internal mechanism **30** includes a connecting member **32**, a resilient member **33**, and a guiding means **34**, and is operatively

connected to the actuator **1** and the pawl **5**, as will be described in detail herein after.

Pawl

[0061] The pawl **5** has a pawl body **5a** having a top portion **5b**, a front portion **5c** facing away the first axis **A1**, a rear portion **5d** opposite the front portion **5c**, and a bottom portion **5e**.

[0062] The pawl body **5a** defines a cavity **5f** configured to receive at least partially the connecting member **32** and the resilient member **33**. As best seen in FIG. **10C**, the cavity **5f** further includes an elongated portion **5j** extending downwardly and away from the first axis **A1**.

[0063] The bottom portion **5e** includes a hook portion **5g** extending in a direction parallel to the third axis **A3** and away from the first axis **A1**, and being configured to be insertable into the pair of openings when the releasable clamp **10** is in the first state **S1** and to apply a clamping force behind the vehicle opening toward the compression surface **4** when the releasable clamp **10** is in the second state **S2**.

[0064] The pawl body **5a** may be provided with a biasing abutment portion **31** within the housing portion **2d** of the base **2** that receives at least partially some of the components of the protrusion **3** that enable operation of the pawl **5**. The biasing abutment portion **31** may be slidable along the guiding means **34** to extend or retract relative to the protrusion **3**, via the internal mechanism **30** pivotally connected to the actuator **1** on a first end thereof about a first pivot and may pivotally connect, on a second end thereof, to the biasing abutment portion **31** of the pawl **5** operated thereby about a second pivot.

[0065] As the actuator **1** is moved in one of the first direction **R1** and the second direction **R2**, the force from the actuator **1** is transferred through the internal mechanism **30** to a pushing or pulling force that pushes or pulls on the biasing abutment portion **31** to slide the biasing abutment portion **31** along the guiding means **34** to extend or retract the pawl **5**. More precisely, as the actuator **1** is activated, the coupler **1c** transfers the operational forces applied to the actuator **1** to the biasing levers **32** in order to urge the pawl **5** into an engaged or disengaged position.

[0066] As can be seen in FIG. **5** and FIG. **10C**, the biasing abutment portion **31** may include a projecting pawl portion **31a** that projects in the longitudinal (insertion) direction, a sloping engagement face **31b**. As the pawl **5** is extended from the protrusion **3**, such as by operation of the actuator **1** as described above, the sloping engagement face **31b** may come into contact with a body portion, more specifically a peripheral portion of the opening of the body portion as will be further described below, (e.g. a vehicle body), to which another body (e.g. an accessory body) is being joined by the releasable clamp **10**. Due to the sloped aspect of the sloping engagement face **31b** of the pawl **5**, the releasable clamp **10** still provides a secure grip notwithstanding the variations of thicknesses among the first and second bodies to be joined. In other words, it can be said that the clamp **10** is configured to provide a clamping force adaptative to various thicknesses of body portions to be joined. The pawl or pawls **5** may also include other features such as a gripping ridge **31c** that may assist in avoiding slippage of the clamp **10** once in operation.

[0067] The pawl body **5a** further has a pawl pivot **5h** defining a pawl pivot axis **5i** extending parallel to the first axis **A1**. In this embodiment, the pawl pivot axis **5i** is movable in the elongated portion **5j** of the cavity **5f** away from the first axis **A1**.

[0068] The pawl body **5a** is operatively connected to the guiding means **24** aligned with the third axis **A3** such that the pawl **5** moves or slides according to a translational movement along the guiding means **24**.

Guiding Means

[0069] In the present embodiment, the guiding means **24** is a circular rod extending parallel to the third axis **A3**. The pawl body **5a** therefore has an opening configured to receive and slide along the guiding means **24**. In other embodiments, the guiding means **24** can be a rail, a notch, or any equivalent linear guiding feature, and the pawl body **5a** can thus be adapted to operatively connect to and slide along said variations of guiding means **24**.

Connecting Member

[0070] The connecting member **32** has an elongated body **32a** having a first end **32b** pivotally connected to the coupler **1c** via the connecting pivot **1d** and about the connecting pivot axis **1e**, and a second end **32c** pivotally connected to the pawl body **5a** via the pawl pivot **5h** and about the pawl pivot axis **5i**.

[0071] In this exemplary embodiment, the connecting member **32** has a first protrusion extending longitudinally upwardly received in a cavity in the coupler **1c** and therefore forming the connecting pivot **1d**, and a second protrusion extending longitudinally downwardly received in the cavity **5f** of the pawl **5** and therefore forming the pawl pivot **5h**. Other equivalent configurations are contemplated to form pivotal connections between the pawl **5**, the connecting member **32**, and the coupler **1c**.

[0072] In some embodiments, the connecting member **32** and the resilient member **33** may be integral and forming a biasing lever. The biasing lever may include a rigid portion **32** and a resilient portion **33**, which, in some embodiments may include a spring.

Resilient Member

[0073] The resilient member **33** is configured to bias the pawl **5** away from the first axis **A1**. The resilient member **33** may be a spring or any equivalent elastic part.

Operation

[0074] In order to operate the releasable clamp **10**, the actuator **1**, which is oriented along the second axis **A2**, may be moved in various directions around the first axis **A1**, such as in a first direction **R1** and in a second direction **R2**, by the application of a force thereto. Moving the actuator **1** around the first axis **A1** results in moving the pawl or pawls **5** along a third axis **A3** extending generally perpendicular to the first axis **A1**, as will be described in greater detail herein after. In some embodiments, **R1** and **R2** may refer to opposite directions of rotation, but the present technology is not so limited. For example, the actuator **1** may move in a single or multiple directions and may engage or disengage the clamp **10** in one or more directions of movement of the actuator **1**. For example, in some embodiments, the operation of the actuator **1** in the first direction **R1** and in the second direction **R2** may enable the releasable clamp **10** to be switched between a disengaged state (first state, **S1**) and an engaged state (second state, **S2**) as will be described in greater detail herein after.

Intermediate State

[0075] As illustrated in FIG. 2, the pawl or pawls **5** being at least partially extended along the third axis **A3**, the releasable clamp **10** may be in the second state or engaged state. It could be said that the clamp **10**, as shown in FIG. 2, is in an intermediate state.

[0076] As the actuator **1** is further engaged for operation, such as by further movement of the actuator **1** in the first direction **R1** or in the second direction **R2**, the first state or second state of the releasable clamp **10** may be further engaged as illustrated in FIG. 3.

[0077] Generally speaking, moving the actuator **1** toward further engaging one state will have the effect of disengaging the other state. That is, as the actuator **1** is moved toward the first state **S1**, the degree of holding or grip provided by the extension pawl or pawls **5** along the third axis **A3** in the second state **S2** is reduced. Conversely, as the actuator **1** is moved toward the second state **S2**, the degree of insertability or removability provided in the first state **S1** is reduced.

State S1

[0078] As illustrated in FIG. 3, the pawl or pawls **5** being fully retracted, the releasable clamp **10** may be considered to be in the first state **S1** or disengaged state, ready for insertion or removal of the clamp **10** into or from the aligned pair of openings. It will be appreciated that the first state or fully retracted state as an initial undeployed state, allows easy insertion or removal of the releasable clamp **10** for operation as will be described in further detail herein after.

[0079] In accordance with one or more embodiments, as illustrated in FIG. 6, the releasable clamp **10** may be used to facilitate the attachment of a first body and a second body. Examples of a first body and a second body may include an accessory body **40** such as a utility caddy, as a first body,

that may be secured in the bed of a vehicle body **41**, as a second body, such as by attachment to a vehicle bed rail. It will be appreciated that, in many applications, the vehicle body **41** may be provided with openings to facilitate the attachment of various accessories such as tie-down straps and other uses. These openings may be used for the attachment of an accessory or other body in accordance with one or more embodiments by being alignable with a corresponding opening provided in the accessory or other body as will be described in greater detail herein after.

[0080] FIG. 7 illustrates an exemplary arrangement in which the releasable clamp **10** may be used to secure the accessory body **40** to the vehicle body **41**. In the illustrated embodiment, an adapter **52** may be used to facilitate use of the releasable clamp **10**. It will be understood that the adapter **52** may provide advantageous effects such as providing a standardized opening **50** for the accessory body **40** specific to the particular vehicle body **41**, such that the accessory body **40** may be used with a variety of vehicle bodies **41** by simply changing the adapter **52**. The adapter **52** may also be used to provide a seating template area for the compression surface **4** of the releasable clamp **10**. The adapter **52** may be placed over one of the holes **51** present in, for example, a bed rail of the vehicle body **41**. The adapter **52** may further be provided with a non-slip surface **52a** placed between the bottom surface of the adapter **52** and the mounting surface of the vehicle body **41** to prevent or reduce rattling and slippage. The adapter **52** may further be provided with fasteners **52b** that may allow the adapter **52** to be adjustably attached to a mounting flange of accessory body **40**. FIG. 7 illustrates releasable clamp **10** in a position for insertion into the hole **50** of the accessory body **40** and the hole **51** of the vehicle body **41**, which are aligned. Prior to insertion, it will be understood that the releasable clamp **10** should be placed in the first state, i.e. the state in which the pawl or pawls **5** are retracted into the protrusion **3** as described above, by applying an operating force to the actuator **1**.

[0081] As can be seen in FIG. 8, once the releasable clamp **10** is placed in the first state, with the pawl or pawls **5** retracted, the releasable clamp **10** may be inserted through the aligned holes **50** and **51** in the accessory body **40** and the vehicle body **41**. After insertion, the releasable clamp **10** may be switched in the second state, wherein the pawl or pawls **5** are extended from the protrusion **3**, by applying an operating force to the actuator **1**, such that the releasable clamp **10** transitions from the first state **S1**, to the intermediate state and to the second state **S2** in which the sloping engagement surfaces **31b** engage with the vehicle body **41**, more specifically with at least a portion of the peripheral portion of the hole **51** and bring the vehicle body **41**, the accessory body **40**, and the compression surface **4** of the releasable clamp **10** into compressed relation. The degree of force with which the engagement surfaces **31b** engage the releasable clamp **10** may be determined by factors such as the combined thicknesses of the accessory body **40** and the vehicle body **41**, the biasing force of the connecting member **32**, the force applied to the actuator **1**, and so on.

[0082] Various views of the releasable clamp **10**, in the second state, wherein the pawl or pawls **5** are extended, are shown in FIG. 9A to FIG. 9E. To better understand the operation of the releasable clamp **10** in subsequent figures, several cross-sectional views are provided and described herein below. In the following figures, the illustrated cross sections may be represented as cross-sections taken along section line AA-AA shown in FIG. 9A for perspective side views, or along section line E-E shown in FIG. 9B for side views. It will be understood that as actuator **1** is moved in the various actuating positions shown in the subsequent figures, the cross section will nevertheless generally represents the cross-section along section line AA such that various cross-sectional views of the actuator **1** are shown in various actuating positions.

[0083] FIG. 10A and FIG. 10B illustrate views of the releasable clamp **10**, such as after being inserted through the hole **50** in accessory body **40** and the hole **51** in the vehicle body **41**, which are aligned, and in the second state wherein the pawl or pawls **5** are extended. More specifically, the releasable clamp **10** is shown in the second state **S2** in which the sloping engagement face **31b** is engaged with the peripheral portion (e.g. edge or edge surface) of the hole **51** of the vehicle body **41**. It will be understood that in order to achieve the second state **S2**, the releasable clamp **10** is in

the first state **S1** or moved from the first state **S1** to the intermediate state by applying a force to the actuator **1** such that the pawl or pawls **5** may be urged laterally into the extended position in which the pawl or pawls **5** are engaged with at least a portion of the peripheral portion of the hole **51** of the vehicle body **41**. The force applied to the actuator **1**, along with the biasing force of connecting member **32** may extend and hold the pawl or pawls **5** in the extended position from the protrusion **3**, notwithstanding the total thickness to be clamped.

[0084] FIG. **10C** to FIG. **10G** illustrate additional views of the releasable clamp **10** and its components when the releasable clamp **10** is in the extended position.

[0085] Referring to FIG. **10E**, when the actuator **1** is rotated from the first position to the second position, the coupler **1c** drives the connecting member **32** away from the first axis **A1** in a translational and rotational movement via the internal mechanism **30** that is operatively connected to the actuator **1** via the connecting pivot **1d**. The connecting member **32** then drives the pawl **5** away from the first axis **A1** in a translational movement guided by the guiding means **34** via the pawl pivot **5h**. When the actuator **1** is rotated such that the coupler **1c** and the connecting member **32** begin to create an over-center effect, the pawl pivot **5h** moves away from the first axis **A1** along the elongated cavity portion **5j**, which activates the resilient member **33**. When the actuator **1** is in the second position, the pawl **5** is moved away from the first axis **A1** along the guiding means **34**. The resilient member **33** provides a force that contributes to maintain the over-center effect that prevents unwanted release of the releasable clamp **10**.

[0086] In some embodiments, as shown in FIG. **10E**, the protrusion **3** can further have stopper(s) (e.g. tab(s), protrusion(s), bump(s), etc.) limiting the movement of certain components (e.g. rotation of the actuator **1** or the connecting member **32**, etc.). This addition helps give feedback as to when the releasable clamp **10** is in the locked position or unlocked position. It can also prevent the mechanism to be damaged by excessive movement or force applied thereto.

[0087] FIG. **11A** to FIG. **11C** illustrate the releasable clamp **10** in an intermediate state, e.g. to a partially extended/retracted position, such as when the pawl or pawls **5** are being moved from the first state **S1** to the second state **S2**, or from the second state **S2** to the first state **S1**. It will be understood that in order to achieve the intermediate state between the first state **S1** and the second state **S2**, the releasable clamp **10** is operated by applying a force to the actuator **1** in a direction **R1** or **R2**. The actuator **1** may be operated such that the pawl or pawls **5** are urged laterally from the extended position, in which the pawl or pawls **5** are fully engaged with the aligned openings, toward the retracted position in which the pawl or pawls **5** disengage from the aligned openings. Alternatively, the actuator **1** may be operated such that the pawl or pawls **5** may be urged laterally from the retracted position, in which the pawl or pawls **5** are fully retracted, toward the extended position in which the pawl or pawls **5** move toward engagement.

[0088] FIG. **12A** and FIG. **12B** illustrate views of the releasable clamp **10**, in an intermediate state **S1'** such as after being placed in the first state **S1** and being inserted through the hole **50** in accessory body **40** and the hole **51** in the vehicle body **41**, which are aligned, and operating the actuator **1** in the direction **R1** or **R2** such that the pawl or pawls **5** begin to extend into the second state **S2**. The intermediate state illustrated in FIG. **12A** and FIG. **12B** may also reflect an intermediate state **S2'** after operating the actuator in the direction **R1** or **R2** to disengage from the second state **S2**, in which the actuator **1** is operated in the direction **R1** or **R2** such that the pawl or pawls **5** begin to retract.

[0089] FIG. **13A** through FIG. **13D** illustrate various views the releasable clamp **10** in the first state **S1** in which the pawl or pawls **5** are fully retracted into the protrusion **3**. It will be understood that in order to achieve the first state **S1**, the releasable clamp **10** is operated by applying a force to the actuator **1** in one of the directions **R1** or **R2** such that the pawl or pawls **5** may be urged laterally into the retracted position in which the pawl or pawls **5** are disengaged. It will be further understood that in order to insert the releasable clamp **10** into operating position, i.e. through holes **50** and **51** of the accessory and vehicle bodies, respectively, the releasable clamp **10** should be in

the first state **S1** and the holes **50** and **51** should be aligned or in a properly registered position. [0090] FIG. **14A** and FIG. **14B** illustrate views of the releasable clamp **10**, in the fully retracted state **S1** such as after being placed in the first state **S1** and being inserted through the hole **50** in accessory body **40** and the hole **51** in the vehicle body **41**, which are aligned, and operating the actuator **1** in the direction **R1** or **R2** such that the pawl or pawls **5** begin to extend outwardly toward the second state **S2**.

Method

[0091] In accordance with one or more embodiments, a method **60** of releasably clamping a first body to a second body is provided as illustrated in FIG. **15**. The method may be practiced, for example, using the releasable clamp **10** as described herein above. In a first step **S61**, the pawl or pawls of the exemplary releasable clamp should be ensured as being in the first state, a state of being housed (retracted) into the base (protrusion) of the clamp. The first and second openings of the respective first and second bodies to be clamped should be brought into alignment or registered. In a second step **S62**, a protrusion of the releasable clamp is inserted through the first and second openings in a longitudinal direction (insertion direction) of the protrusion. When the releasable clamp is fully seated into the openings, the clamping surface of the releasable clamp is resting on an outer surface of the first body.

[0092] In a third step **S63**, the state of the pawl or pawls is switched to the second state in which the pawl or pawls extend from the base (protrusion) in a lateral direction, perpendicular to the longitudinal direction. The second state may be achieved in response to power transmitted by applying an operational force to an actuator of the clamp. In a fourth step **S64**, as the second state is achieved, and the pawl or pawls move to an engaged position, the second body, the first body and the base are brought into a compressed relation against a compression surface of the releasable clamp that is in contact with a surface of the first body such that the first body and the second body are sandwiched between the compression surface and the sloping engagement surfaces of the pawl or pawls.

Effects

[0093] One or more embodiments of the releasable clamp **10** and method of releasably joining first and second body described herein provide distinct advantages. For example, due to the sloping engagement surfaces of the pawl or pawls **5**, varying thicknesses of the first body and the second body may be taken into account without the need for specialized designs for each combination of body thicknesses. To the contrary, one or more embodiments as described herein may accommodate a wide variety of thicknesses and still provide reliable and rapid clamping and releasing.

[0094] More precisely, when the sloping engagement surface **31b** comes into contact with a body portion, more specifically a peripheral portion of the opening of the body portion as will be further described below, (e.g. a vehicle body), to which another body (e.g. an accessory body) is being joined by the releasable clamp **10**, when the actuator **1** is rotated from the first position to the second position, the internal mechanism **30** drives the pawl **5** away from the first axis **A1** in a translational movement guided by the guiding means **34** via the pawl pivot **5h**, such that the pawl pivot **5h** moves away from the first axis **A1** along the elongated cavity portion **5j**, which activates the resilient member **33**. When the actuator **1** is in the second position, the pawl **5** is moved away from the first axis **A1** along the guiding means **34** in locked position. In other words, the locking process (i.e. movement of the pawl pivot, compression of the resilient member, creation of an over-center effect, etc.) of the releasable clamp **10** begins when the sloping engagement surface **31b** comes into contact with a peripheral portion of the opening into which the releasable clamp **10** is engaged, notwithstanding the thickness of the body portion to clamp. It can thus be said that the releasable clamp **10** is configured to provide a clamping force adaptative to various thicknesses of body portions to be joined, thanks to the internal mechanism and the sloping engaging portion, among other things.

[0095] The present technology may include other embodiments or modifications in addition to one or more embodiments and modifications described herein without departing from the spirit of the invention. The one or more embodiments and modifications described herein are to be considered illustrative and exemplary, and not restrictive, exhausting or limiting in nature. The scope of the present technology is indicated by the appended claims rather than by the foregoing description. Accordingly, configurations considered to fall within equivalent arrangements of the claims are intended to be embraced within the proper scope of the invention.

Claims

1. A clamp for releasably joining a first body to a second body, the clamp being insertable through a first opening in the first body and a second opening in the second body aligned with the first opening, the clamp comprising: a base comprising a compression surface that faces the first body and is compressible around a periphery of the first opening, and a protrusion extending in a longitudinal direction from the base, the protrusion insertable through the first and second openings and connectable to the second body; an actuator connected to the base; and a retractable pawl assembly housed in the base, the pawl assembly having a pawl or pawls extendable from the protrusion of the base in a lateral direction perpendicular to the longitudinal direction, the pawl assembly operatively coupled to the actuator, wherein the pawl or pawls are retracted into the protrusion in a disengaged position by operation of the actuator through a force applied to the actuator, and the pawl or pawls are extended from the protrusion of the base to an engaged position to engage the second body and to hold the second body, the first body and the base into compressed relation against the compression surface by force applied to the actuator.
2. The clamp according to claim 1, wherein each of the pawl or pawls comprises a cam surface that slopes to widen in a direction away from the base to variably engage with the second body.
3. The clamp according to claim 2, wherein the cam surface extends to variably engage the second body to hold the second body, the first body and the base into compressed relation against the compression surface.
4. The clamp according to claim 1, wherein the base further comprises resilient members coupled to the actuator at one end and extending laterally from a central axis of the base in the longitudinal direction, the resilient members pressing against the pawl or pawls at an other end to urge the pawl or pawls to extend laterally outwardly by operation of the actuator.
5. The clamp according to claim 1, wherein the pawl or pawls extend from and retract into the protrusion in a direction orthogonal to the rotation axis.
6. A clamp for releasably joining a first body to a second body, the clamp being insertable through a first opening in the first body and a second opening in the second body aligned with the first opening, the clamp comprising: a base having a compression surface; a protrusion projecting from the base in a longitudinal direction perpendicular to the compression surface relative to the base; a retractable pawl that can be switched between a first state of being housed in the protrusion and a second state of extending from the protrusion in a lateral direction perpendicular to the longitudinal direction; and an actuator that transmits power to the retractable pawl for switching the state of the retractable pawl from one of the first and second states to the other one of the first and second states in response to an operation force; wherein the retractable pawl comprises a surface facing the base and that slopes away from the compression surface in the longitudinal direction.
7. The clamp according to claim 6, wherein the surface of the pawl widens in a lateral direction away from the base.
8. The clamp according to claim 7, wherein the surface of the pawl widens in the lateral direction away from the base to engage with the opening of the second body to hold the second body against the first body, the first body and the base into compressed relation against the compression surface.
9. The clamp according to claim 6, wherein the base further comprises a resilient member

operatively coupled to the pawl at one end and extending laterally from the protrusion to press against the pawl at an other end to urge the pawl outwardly by the operational force.

10. The clamp according to claim 6, wherein the laterally retractable pawl extends laterally from and retracts laterally into the protrusion in a direction orthogonal to the longitudinal direction.

11. A method of releasably joining a first body to a second body using a clamp comprising: a base having a compression surface; a protrusion projecting from the base in a longitudinal direction perpendicular to the compression surface relative to the base; a retractable pawl retractable into the base and comprising a surface; and an actuator that transmits power to the retractable pawl, the method comprising: inserting, with the retractable pawl in a first state of being retracted, the protrusion through a first opening in the first body and a second opening in the second body aligned with the first opening; and switching the state of the retractable pawl to a second state of being extended from the base in a lateral direction perpendicular to the longitudinal direction in response to the power transmitted by applying an operational force applied to the actuator; wherein in the second state, the retractable pawl extends from the base to an engaged position to engage the second body and to hold the second body, the first body and the base into compressed relation against the compression surface.

12. The method of claim 11, further comprising switching the state of the retractable pawl from the second state in which the second body, the first body and the base are held in compressed relation against the compression surface to the first state in which the second body, the first body and the base are released from compressed relation such that the clamp is removed.

13. An actuator for engaging a releasable clamp for releasably joining a first body to a second body, the releasable clamp being insertable through a first opening in the first body and a second opening in the second body aligned with the first opening, the releasable clamp comprising: a base; a protrusion; a retractable pawl extendable and retractable along an engagement axis, the retractable pawl capable of being switched between a first state of being housed in the protrusion and a second state of extending from the protrusion, the actuator comprising: an actuator body configured to be selectively rotatable about a first axis extending longitudinally, the actuator body extending along a second axis perpendicular to the first axis, the actuator body configured to activate operation of the releasable clamp under an activation force; a mounting portion operatively coupled to the base; a coupler operatively coupled to the actuator and extending radially from the first axis; and an internal mechanism operatively coupled to the actuator and the retractable pawl, the internal mechanism configured to convert the rotational movement of the actuator under the activation force about the first axis into a translational movement of the retractable pawl along the engagement axis, the internal mechanism comprising: a connecting member pivotally connected to the coupler about a connecting pivot defining a connecting pivot axis extending parallel to the first axis, a resilient member provided between the connecting member and the pawl to bias the pawl away from the first axis, and a guiding member slidingly connected to and guiding the connecting member along the engagement axis.

14. The actuator according to claim 13, wherein the actuator is configured to transmit the activation force to the retractable pawl through the internal mechanism, the actuator for switching the state of the retractable pawl from one of a first state in which the pawl is extended and a second state in which the pawl is retracted to the other one of the first state and the second state in response to the activation force; and the retractable pawl comprises an engagement surface facing the base and that slopes away in the longitudinal direction from a compression surface of the base.

15. The actuator according to claim 14, wherein the engagement surface of the retractable pawl widens in a lateral direction away from the base.

16. The actuator according to claim 14, wherein when the engagement surface comes into contact with a peripheral portion of the first and second openings of the first and second bodies, respectively, the releasable clamp provides a clamping force adaptative to various thicknesses among the first and second bodies to be joined.

17. The actuator according to claim 14, wherein when the activation force is applied to the actuator such that the retractable pawl is in the second state, the resilient member applies the biasing force so as to maintain an over-center effect between the coupler and the connecting member that prevents unwanted release of the releasable clamp.

18. The actuator according to any one of claim 13, wherein the actuator is operable without use of a tool.

19. A releasable clamp for releasably joining a first body to a second body, the releasable clamp being insertable through a first opening in the first body and a second opening in the second body aligned with the first opening, the releasable clamp comprising: a base; a protrusion; a retractable pawl extendable and retractable along an engagement axis, the retractable pawl capable of being switched between a first state of being housed in the protrusion and a second state of extending from the protrusion; and an actuator in accordance with claim 13, wherein the actuator is configured to transmit the activation force to the retractable pawl through the internal mechanism so as to switch the state of the retractable pawl from one of the first state and the second state to the other one of the first state and the second state in response to the activation force.
